Repeated prime factors and strong three-dimensional partitions

Human Compression project: supplementary proof for F008

Theorem 1.3 of Lebowitz–Lockard and Tenenbaum [1] concerns integral partitions of a cube into equal-volume boxes whose lengths along one fixed axis are pairwise distinct. We prove that the set of side lengths admitting such a partition has an asymptotic density strictly less than one. This supplements the already reviewed complete companions *Divisor concentration and primitive bases* and *Primitive rectangular partitions and planar density*. We acquire both companions in full; no part is treated as a free earlier argument.

The source's primitive-basis and prime-divisor strategy is retained and attributed. The new argument replaces the divisor decomposition and Fourier estimates in its Section 6. In particular, we impose a stronger small-prime cutoff and assign each divisor its *largest repeated large prime*. That assignment makes the two-prime constraints independent. No finite calculation is a premise of this uniform proof.

Write $\log_2 x = \log\log x$. All constants below are independent of x, which is sufficiently large. We use Chebyshev's bound $\pi(u) \ll u/\log u$ and Euler summation for the zeta function, already retained in the companions. The parameters in the final application are fixed as

$$\varepsilon = \frac{1}{100}, \qquad c = \frac{49}{50}, \qquad \kappa = \frac{21}{20}.$$

1 A cutoff that forces a repeated large prime

Put $L = \log x$, $\xi = \log L$, and

$$z = \exp(L/\xi^2), \qquad Y = \exp(L/(10\xi)), \qquad n_z = \prod_{\substack{\ell^a \| n \\ \ell \le z}} \ell^a.$$

Here and below letters p, q, ℓ used as prime factors denote primes.

Lemma 1 (Small-prime part)**.** *For every fixed $A > 0$,*

$$\#\{n \le x : n_z > Y\} \ll_A xL^{-A}.$$

Proof. Set $\lambda = \log\xi/\log z$; eventually $0 < \lambda < 1/4$. The multiplicative function $f(n) = n_z^\lambda$ has nonnegative divisor-sum coefficients: at $\ell \le z$ these are $g(\ell^a) = \ell^{a\lambda} - \ell^{(a-1)\lambda}$ for $a \ge 1$, and at $\ell > z$ they are zero. Consequently

$$\sum_{n\le x} f(n) \le x \sum_{d\ge 1} \frac{g(d)}{d} = x \prod_{\ell \le z} \frac{1 - 1/\ell}{1 - \ell^{\lambda-1}}.$$

The logarithm of this product is

$$\sum_{\ell \le z} \frac{\ell^\lambda - 1}{\ell} + O(1),$$

uniformly for $\lambda < 1/4$, since all terms of degree at least two are bounded by a constant times $\sum_\ell \ell^{-3/2}$. Chebyshev's bound and partial summation give

$$\sum_{\ell \le z} \frac{\ell^\lambda - 1}{\ell} \ll 1 + \frac{z^\lambda}{\log z} + \int_2^z \frac{(t^\lambda - 1) + \lambda t^\lambda}{t \log t}\, dt \ll 1 + z^\lambda.$$

For the last estimate substitute $v = \lambda \log t$. On $v \le 1$, $(e^v - 1)/v$ is bounded and $\lambda \int_{\lambda \log 2}^{1} v^{-1}\, dv = O(1)$; on $v \ge 1$ both integrals are $O(e^{\lambda \log z})$. As $z^\lambda = \xi$, Markov's inequality now gives

$$\#\{n \le x : n_z > Y\} \le x \exp\{-\lambda \log Y + O(\xi)\} = x \exp\{-\tfrac{1}{10}\xi \log \xi + O(\xi)\}.$$

This is $O_A(xL^{-A})$ for every fixed A. □

Call b *good* if

$$\sqrt{x} < b \le x, \quad \Omega(b) \le (1+\varepsilon)\log_2 b, \quad b_z \le Y, \quad \ell^2 \mid b \Longrightarrow \ell \le z. \tag{1}$$

The last two conditions discard at most $O_A(xL^{-A})$ integers from the first two: use Lemma 1 and $x \sum_{j>z} j^{-2} \ll x/z$.

Lemma 2 (Canonical repeated prime)**.** *Let b be good, $p = P^+(b)$, and $m = b/p$. Then $p \nmid m$,*

$$p \ge x^{1/(2(1+\varepsilon)\xi)} > z, \qquad \tau(b^3) \le 4^{\Omega(b)} \le L^{(1+\varepsilon)\log 4} < L^2.$$

Every divisor $d \mid m^3$ in $[b/L^2, b)$ has a prime $q > z$ whose exponent in d is two or three. Thus it has a well-defined largest such prime, denoted $q(d)$. If $q(d_1) > q(d_2)$, the exponent of $q(d_1)$ in d_2 is at most one.

Proof. Since $b \le p^{\Omega(b)}$ and $b > \sqrt{x}$, the lower bound for p follows. For large x it exceeds z, so p occurs to exponent one by (1). The divisor bound follows from $3a + 1 \le 4^a$.

If a divisor $d \mid m^3$ had exponent at most one at every prime above z, then

$$d \le b_z^3 \frac{m}{b_z} = b_z^2 m \le \frac{Y^2 b}{p} < \frac{b}{L^2}.$$

The last inequality holds because $\log(Y^2 L^2) = L/(5\xi) + 2\xi$ is smaller than $L/(2(1+\varepsilon)\xi)$. This proves existence; the final assertion is the definition of the largest repeated large prime. □

2 A uniform weighted divisor-pair estimate

For real θ, put $\tau(v, \theta) = \sum_{d \mid v} d^{i\theta}$.

Lemma 3 (Fourier mean)**.** *For fixed $0 < y \le 1$, uniformly for $x \ge 3$,*

$$\int_0^1 \sum_{h \le x} \frac{y^{\Omega(h)} |\tau(h^3, \theta)|^2}{h}\, d\theta \ll_y L^{4y + (12y-1)^+} \begin{cases} 1 + \log L, & 12y = 1, \\ 1, & 12y \ne 1. \end{cases}$$

In particular, at $y = 1/12$ the bound is $O(L^{1/3}(1+\xi))$.

Proof. Take $\sigma = 1 + 1/L$. Nonnegativity permits replacing the truncated harmonic sum by at most e times its full Dirichlet series $F(\sigma, \theta)$. It is an Euler product, since $y^{\Omega(h)}|\tau(h^3, \theta)|^2$ is multiplicative. Its terms at a prime power ℓ^a are at most $y^a (3a+1)^2 \ell^{-a\sigma}$; those with $a \ge 2$ have uniformly bounded total contribution to the logarithm.

The exact first-degree term is determined by

$$|1 + e^{iu} + e^{2iu} + e^{3iu}|^2 = 4 + 6\cos u + 4\cos(2u) + 2\cos(3u). \tag{2}$$

Indeed each coefficient counts ordered exponent pairs of a specified difference. Using the absolutely convergent logarithm of the Euler product for ζ, whose prime-power terms of degree at least two are uniformly bounded, we obtain

$$F(\sigma, \theta) \ll_y \zeta(\sigma)^{4y} |\zeta(\sigma + i\theta)|^{6y} |\zeta(\sigma + 2i\theta)|^{4y} |\zeta(\sigma + 3i\theta)|^{2y}.$$

This argument uses $\log(1+u) \le u$ on each positive local factor; no assumption on zeros on the line $\Re s = 1$ is required.

Euler summation gives, for $\Re s > 1$,

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{v\} v^{-s-1}\, dv.$$

For $1 < \Re s \le 2$ and $|\Im s| \le 3$ this implies $|\zeta(s)| \ll 1 + |s-1|^{-1}$. Therefore, for $0 < \theta \le 1$ and $j = 1, 2, 3$, $|\zeta(\sigma + ij\theta)| \ll \min(L, 1/\theta)$, while $\zeta(\sigma) \ll L$. Integrating $L^{4y}\min(L, 1/\theta)^{12y}$ over $0 < \theta \le 1$ proves all three cases of the stated bound. □

Lemma 4 (Shifted pair concentration)**.** *Let $a_1, a_2 \in \{2, 3\}$, $V \in \mathbb{R}$, and $W \ge 3$. Define*

$$D(h) = \#\{(e_1, e_2) : e_1, e_2 \mid h^3, |a_1 \log e_1 - a_2 \log e_2 - V| \le W\}.$$

Uniformly in V, for $W \ll \xi$ and x large,

$$\sum_{\substack{h \le x \\ \Omega(h) \le (1+\varepsilon)\xi}} \frac{D(h)}{h} \ll \xi^2 L^{1/3 + (1+\varepsilon)\log 12}.$$

Proof. On $|v| \le W$ the function $\big(\sin(v/(2W))/(v/(2W))\big)^2$ is bounded below by a positive absolute constant. Its Fourier representation is

$$\left(\frac{\sin(v/(2W))}{v/(2W)}\right)^2 = W\int_{-1/W}^{1/W} (1 - W|\theta|) e^{i\theta v}\, d\theta.$$

Summing this majorant, taking absolute values, and using $|AB| \le (|A|^2 + |B|^2)/2$ show

$$D(h) \ll W\int_0^1 |\tau(h^3, u)|^2\, du.$$

The substitutions $u = a_j\theta$ are valid because $a_j/W \le 1$; the phase $e^{-iV\theta}$ has modulus one. For $y = 1/12$, the restriction on $\Omega(h)$ gives $1 \le y^{\Omega(h)} L^{-(1+\varepsilon)\log y}$. Apply Lemma 3, and use $W \ll \xi$. □

Lemma 5 (Prime windows)**.** *For $z \ge 3$ and a real interval J of length $W \ge 0$,*

$$\sum_{\substack{q > z \\ \log q \in J}} \frac{1}{q} \ll \frac{W+1}{\log z}.$$

If primes $p > z$ are additionally restricted by $p \le T$, their number is at most $O(T/\log z)$, with zero possibilities when $T \le z$.

Proof. Intersect the interval with $[\log z, \infty)$ and divide it into at most $W+2$ intervals of length at most one. In such an interval $[u, u+1]$, Chebyshev's bound gives $\sum_{e^u < q \le e^{u+1}} q^{-1} \le e^{-u}\pi(e^{u+1}) \ll 1/u \le 1/\log z$. The second assertion is Chebyshev's bound at $T > z$. □

3 The two repeated-prime cases

For good $b = mp$ put

$$T(b) = \#\{d \mid m^3 : b/L^2 \le d < b\}, \qquad \alpha_1 = (1+\varepsilon)\log 12 - \frac{5}{3}, \quad \alpha_2 = (1+\varepsilon)\log 16 - 2.$$

Proposition 6 (Second moment near the largest prime)**.** *For $\sqrt{x} \le X \le x$,*

$$\sum_{\substack{X < b \le \min(2X, x) \\ b \ mgood}} T(b)^2 \ll X\xi^8\big(L^{\alpha_1} + L^{\alpha_2}\big).$$

Proof. Expand the square into ordered divisor pairs (d_1, d_2) and use the assignment of Lemma 2. Every divisor in the sum lies in $[X/L^2, 2X)$.

Equal assigned primes. Suppose $q(d_1) = q(d_2) = q$. Write $b = hqp$ and $d_j = e_j q^{\nu_j}$, where $e_j \mid h^3$ and $\nu_j \in \{2, 3\}$. This is legitimate because all primes above z occur to exponent one in b, and $q < p$. For fixed h, q, there are at most $O(X/(hq \log z))$ possible primes p. The restrictions on each divisor imply

$$\log X - 2\xi \le \log e_j + \nu_j \log q < \log X + \log 2.$$

Their simultaneous validity implies

$$|\nu_2 \log e_1 - \nu_1 \log e_2 - (\nu_2 - \nu_1) \log X| \le 20\xi.$$

For such a pair (e_1, e_2), the allowed $\log q$ belong to an interval of length $O(\xi)$, independently of p. Lemma 5 and then Lemma 4 give an upper bound

$$\frac{X\xi}{(\log z)^2} \sum_{\substack{h \le x \\ \Omega(h) \le (1+\varepsilon)\xi}} \frac{D(h)}{h} \ll X\xi^7 L^{-2+1/3+(1+\varepsilon)\log 12} = X\xi^7 L^{\alpha_1}.$$

There are only four choices for (ν_1, ν_2). Coprimality and ordering restrictions may safely be discarded in these upper bounds. The shifted pair estimate is needed also when $\nu_1 \ne \nu_2$; ordinary unshifted closeness of e_1, e_2 is not assumed.

Distinct assigned primes. By interchanging the two divisors at a cost of at most two, suppose $q_1 = q(d_1) > q_2 = q(d_2)$. Write $b = hq_1q_2p$ and

$$d_1 = e_1 q_1^{\nu_1} q_2^{\mu_1}, \qquad d_2 = e_2 q_1^{\mu_2} q_2^{\nu_2},$$

where $e_1, e_2 \mid h^3$, $2 \le \nu_1, \nu_2 \le 3$, $0 \le \mu_1 \le 3$, and $0 \le \mu_2 \le 1$. The last inequality is exactly the largest-prime rule. In particular

$$\det \begin{pmatrix} \nu_1 & \mu_1 \\ \mu_2 & \nu_2 \end{pmatrix} = \nu_1\nu_2 - \mu_1\mu_2 \ge 4 - 3 = 1.$$

For fixed h, e_1, e_2 and these exponents, the two logarithmic divisor restrictions define a parallelogram for $(u, v) = (\log q_1, \log q_2)$. Each defining interval has length $2\xi + \log 2$. The inverse matrix has uniformly bounded coefficients, so each coordinate projection of the parallelogram has length $O(\xi)$, with endpoints independent of p.

For fixed q_1, q_2, there are $O(X/(hq_1q_2 \log z))$ possible p. Bounding the parallelogram by its coordinate projections and using Lemma 5 twice gives the total upper bound

$$\frac{X\xi^2}{(\log z)^3} \sum_{\substack{h \le x \\ \Omega(h) \le (1+\varepsilon)\xi}} \frac{\tau(h^3)^2}{h}.$$

Since $\tau(h^3)^2 \le 16^{\Omega(h)} \le L^{(1+\varepsilon)\log 16}$ and $\sum_{h \le x} h^{-1} \ll L$, this is $O(X\xi^8 L^{(1+\varepsilon)\log 16 - 2})$. There are only finitely many exponent choices. The two cases exhaust all ordered divisor pairs and prove the proposition. □

Theorem 7 (Exceptional divisors)**.** *For the fixed constants above,*

$$\#\{b \ mgood : T(b) > L^c\} \ll xL^{-\kappa}.$$

Proof. Apply Proposition 6 to dyadic intervals covering $(\sqrt{x}, x]$. Their left endpoints have sum $O(x)$; thus there is no additional factor $\log x$ in the resulting bound. Markov's inequality gives

$$\#\{b \ mgood : T(b) > L^c\} \ll x\xi^8 \big(L^{\alpha_1 - 2c} + L^{\alpha_2 - 2c}\big).$$

For an entirely rational check of the strict margin, the exponential series through degree eight gives $e^{5/2} > 12$ and $e^{14/5} > 16$. Thus $\log 12 < 5/2$, $\log 16 < 14/5$, and

$$\alpha_1 < \frac{103}{120}, \qquad \alpha_2 < \frac{207}{250} < \frac{103}{120}, \qquad 2c - \frac{103}{120} - \frac{21}{20} = \frac{31}{600} > 0.$$

For reference, the sharper values are

$$\alpha_1 = 0.8430890496\ldots, \qquad \alpha_2 = 0.8003146094\ldots,$$

and $2c - \max(\alpha_1, \alpha_2) = 1.1169109503\ldots$. The fixed factor ξ^8 is absorbed by the rational strict margin above; the decimals are explanatory. □

4 The primitive basis and density

A *strong three-dimensional partition* has at least two integral axis-parallel boxes of equal volume and has pairwise distinct lengths along one specified axis. Let B be the set of side lengths admitting such a partition with no smaller divisor admitting one.

Theorem 8. *The set of strong three-dimensional Mondrian side lengths has an asymptotic density strictly less than one. In particular, the conclusion of source Theorem 1.3 holds.*

Proof. We explicitly use three complete earlier arguments. The companion common paper proves that the multiples of integers b with $\Omega(b) > (1+\varepsilon)\log_2 b$ have vanishing upper-density tails. It also proves the finite Behrend inequality and the consequence that a set of multiples has density less than one when its primitive basis excludes one and has vanishing upper-density tails. The companion application proves its geometric trace lemma and the "few sums, few prime factors" counting lemma for bounded nonnegative coefficients. These statements concern arbitrary dimension and are used with $k = 3$.

Split $B = B_1 \cup B_2$ according as the displayed prime-factor bound fails or holds. The common tail lemma handles B_1. We show $\#(B_2 \cap [1, x]) \ll xL^{-\kappa}$, so the reciprocal sum over B_2 converges. Integers at most $\sqrt{x}$ are harmless. Among the remaining integers of B_2, the nongood ones are $O_A(xL^{-A})$ by Lemma 1 and the large prime-square bound. Theorem 7 discards $O(xL^{-\kappa})$ further integers and leaves $T(b) \le L^c$.

For each remaining $b = mp$, the prime p is a proper divisor: the companion geometric lemma proves that a prime side length has no strong partition. Primitivity therefore supplies a coordinate trace of length b with at least one segment not divisible by p. Every segment length divides b^3, is at least $b/\tau(b^3) \ge b/L^2$, and is less than b. Let D be the sum of the segments not divisible by p. Then $1 \le D \le b \le x$, and reducing the trace modulo p gives $p \mid D$. All its distinct summands divide m^3 and are among the at most $T(b) \le L^c$ possibilities. Each multiplicity is at most the total number of boxes, which is at most $\tau(b^3) \le L^2$. The chosen trace need not be along the axis with distinct lengths; this is precisely why bounded multiplicities are retained.

Moreover $p \ge x^{1/(2(1+\varepsilon)\xi)} > x^{1/(4\xi)}$ and $\tau(m^3) \le L^2$. The complete earlier counting lemma, with $k = 3$, $C = 2$, and $c < 1$, therefore bounds the remaining integers by $O_A(xL^{-A})$ for every fixed A. This proves the claimed bound for B_2.

Partial summation gives $\sum_{b \in B_2} 1/b < \infty$, and hence its multiples have vanishing upper-density tails. The same holds for B_1. Scaling partitions and taking a least divisor with a partition identify all strong Mondrian side lengths with the set of multiples of B. Since $1 \notin B$, the acquired Behrend-tail argument completes the proof, including existence of the density. □

5 Exact source boundary

This proof recovers the uniform conclusion of Theorem 1.3, not the literal intermediate estimates of Section 6. The source's displayed identity $|\tau(\ell^3, \theta)|^2 = 16\cos^6(\theta \log \ell/2)$ is false: at $\theta \log \ell = \pi/2$ its two sides are zero and two. Equation (2) is the exact replacement.

Also, knowing that a divisor has a prime above the cutoff does not imply that such a prime occurs with exponent at least two. We make no claim that the source's particular set S satisfies our stronger assertion. Lemma 1 changes the cutoff and proves an additional negligible-exception condition; Lemma 2 then establishes the assertion actually needed. In the equal-prime case we use two prime-counting logarithmic savings, not an unjustified third saving. In the distinct-prime case the nonsingular exponent matrix is proved from the canonical assignment, not assumed. The corrected geometric trace likewise keeps multiplicities on an arbitrary chosen axis.

No Mondrian partition is exhibited or ruled out individually by a finite search. Conjecture 1.1, the proposed higher-dimensional extension, and the weaker condition that only the boxes themselves be distinct remain unanswered. This supplement does not establish the additional general multiplicative-weight or improved-error assertions in the source's Section 4 remarks. Those boundaries must remain explicit in any subsequent whole-source assembly or comparison.

References

[1] N. Lebowitz-Lockard and G. Tenenbaum, *On solutions to the Mondrian Puzzle*, retained preprint dated 28 August 2026, 13 pages, especially Section 6, pp. 10–12.

[2] Human Compression project, *Divisor concentration and primitive bases* and *Primitive rectangular partitions and planar density*, accepted bounded F008 planar packet and its distinct independent review.